Transformation In ETL Assignment

Q1. Define Data Transformation In ETL And Explain Why It Is Important.

Ans.

Data Transformation Is The T In ETL. It Means Converting Raw Data Into A Clean And Proper Format Before Loading It Into The Data Warehouse.

Why It Is Important:

Data Quality - It Removes Errors, Duplicates, And Empty Values From Data.

Consistency - It Makes Sure All Data Follows The Same Format Across Sources.

Business Rules - It Applies Logic Like Calculations And Categorization.

Performance - Processed Data Makes Queries Run Faster.

Security - Sensitive Data Is Masked To Meet Compliance Rules.

Q2. List Any Four Common Activities Involved In Data Cleaning.

Ans.

1. Handling Missing Values - Filling Or Removing Blank Fields In The Dataset.

2. Removing Duplicates - Deleting Repeated Rows Or Records.

3. Fixing Inconsistent Formats - Standardizing Date, Phone, Gender Entries Etc.

4. Outlier Detection - Finding And Handling Extreme Or Incorrect Values.

Q3. What Is The Difference Between Normalization And Standardization?

Ans.

Normalization:

It Scales Data Between 0 And 1 Using The Formula: (X - Min) / (Max - Min). It Is Also Called Min-Max Scaling And Is Used When There Are No Outliers In The Data.

Standardization:

It Converts Data To Have Mean = 0 And Std = 1 Using The Formula: (X - Mean) / Std Deviation. It Is Also Called Z-Score And Is Used When Outliers Exist In The Data.

Q4. A Dataset Has Missing Values In The Age Column. Suggest Two Techniques To Handle This And Explain When They Should Be Used.

Ans.

Technique 1 - Mean/Median Imputation:

Replace Missing Age Values With The Mean Or Median. Use This When Missing Values Are Few And Random. Use Median When Outliers Are Present.

Technique 2 - Row Deletion:

Delete The Rows Where Age Is Missing. Use This Only When Very Few Rows Are Affected (Less Than 5%) And It Will Not Impact The Analysis.

Q5. Convert The Following Inconsistent Gender Entries Into A Standardized Format (Male, Female): ["M", "Male", "F", "Female", "MALE", "F"]

Ans.

"M" -> Male

"Male" -> Male

"F" -> Female

"Female" -> Female

"MALE" -> Male

"F" -> Female

All Entries Are Converted By Checking The First Letter. M Means Male And F Means Female.

Q6. What Is One-Hot Encoding? Give An Example With The Categories: Red, Blue, Green.

Ans.

One-Hot Encoding Converts Categorical Data Into Numbers. Each Category Gets Its Own Column With Value 1 If Present Or 0 If Not.

Q7. Explain The Difference Between Data Integration And Data Mapping In ETL.

Ans.

Data Integration:

It Is The Process Of Combining Data From Multiple Sources Like CRM, ERP, Excel Into One Single Dataset. It Answers Where The Data Comes From.

Data Mapping:

It Is The Process Of Linking Each Field From The Source To A Field In The Target. It Answers How The Fields Are Connected. Example: Source Column Cust_Name Maps To Target Column Customer_Name.

Q8. Explain Why Z-Score Standardization Is Preferred Over Min-Max Scaling When Outliers Exist.

Ans.

Min-Max Scaling Uses Min And Max Values In The Formula. When Outliers Are Present, The Min And Max Become Extreme Which Squishes All Other Values Into A Very Narrow Range.

Z-Score Uses Mean And Standard Deviation. Since It Does Not Depend On Min Or Max, Outliers Have Much Less Effect On The Result.

So Z-Score Is Preferred When Outliers Exist Because It Gives A More Accurate And Reliable Scaling Of Data.